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Form Approved
OMB No 0704-0188

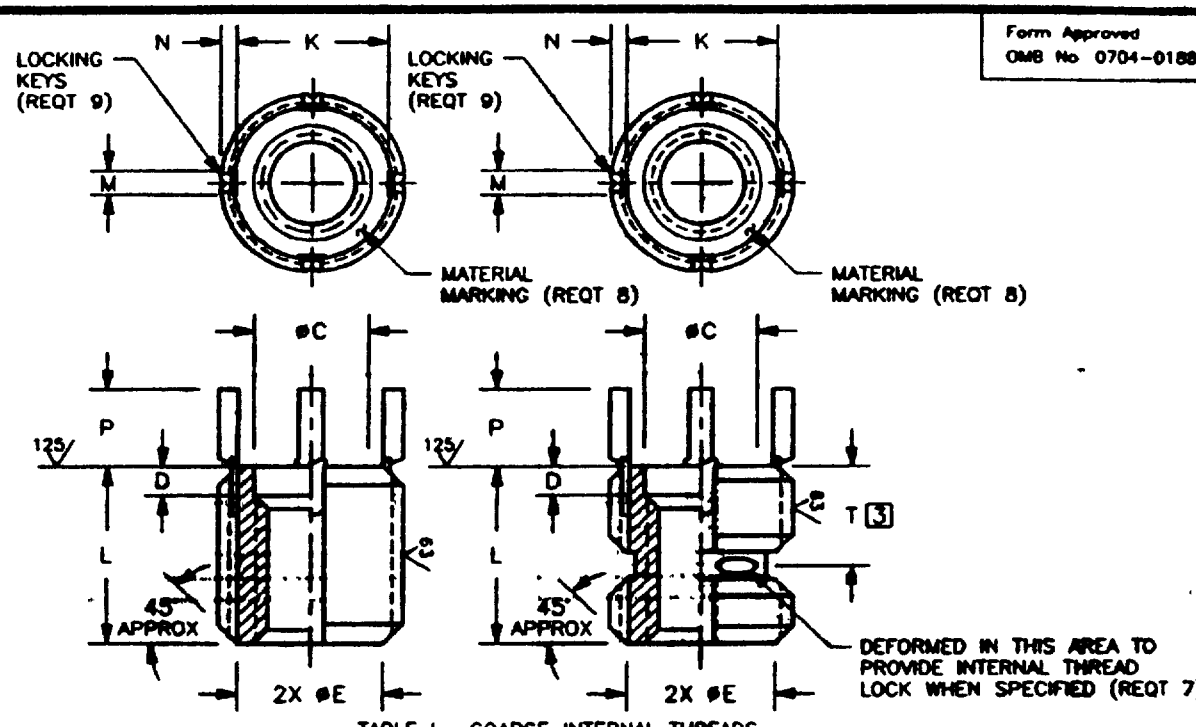



TABLE I COARSE INTERNAL THREADS

| DASH NO | INTERNAL THREAD MIL-S-8879 UNJC-3B | EXTERNAL THREAD FED-STD-H28/2 EXCEPT MODIFIED MINOR DIA | | øC | D | øE | K | L | KEY DIMENSIONS | | | T |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SIZE | øMINOR | ±.005 | ±.010 | REF | REF | ±.015 | M REF | N REF | P REF | ±.030 |
| 101 | .190-24 | .375-16UNC-2A | .320<br>.312 | .196 | .070 | .310 | .284 | .312 | .068 | .040 | .130 | .220 |
| 102 | .250-20 | .4375-14UNC-2A | .383<br>.375 | .257 | .070 | .375 | .344 | .375 | .068 | .040 | .190 | .220 |
| 103 | .3125-18 | .500-13UNC-2A | .440<br>.432 | .316 | .070 | .430 | .407 | 437 | .068 | .040 | 190 | .220 |
| 104 | .375-16 | .5625-12UNC-2A | .503<br>495 | .380 | .070 | .495 | .469 | .500 | .068 | .040 | .190 | .250 |
| 105 | .4375-14 | .625-11UNC-2A | .585<br>.551 | 445 | 100 | 550 | .532 | .625 | .068 | .040 | .190 | .280 |
| 106 | .500-13 | .6875-11NS-2A | .625<br>.615 | .507 | .100 | .615 | .595 | 688 | 068 | .040 | .190 | .280 |
| 107 | .5625-12 | .8125-16UN-2A | .752<br>.744 | .580 | .100 | .745 | .720 | .812 | .068 | 040 | .190 | .320 |
| 108 | .625-11 | .875-14UNF-2A | .815<br>.807 | .640 | .100 | .805 | .782 | .875 | .068 | .040 | .190 | .330 |
| 109 | .750-10 | 1.125-12UNF-2A | 1.050<br>1.040 | .763 | .100 | 1.040 | .975 | 1.125 | .103 | .070 | .250 | --- |
| 109L | .750-10 | 1.125-12UNF-2A | 1.050<br>1.040 | .763 | .100 | 1.040 | .975 | 1.250 | .103 | .070 | .250 | .490 |
| 110 | .875-9 | 1.250-12UNF-2A | 1.175<br>1.165 | .888 | .100 | 1.165 | 1.100 | 1.250 | .103 | .070 | .250 | --- |
| 110L | 875-9 | 1.250-12UNF-2A | 1.175<br>1.165 | 888 | .100 | 1.165 | 1.100 | 1.375 | .103 | .070 | .250 | .490 |
| 111 | 1.000-8 | 1 375-12UNF-2A | 1.300<br>1.290 | 1.015 | .100 | 1.290 | 1.225 | 1.375 | .103 | .070 | .250 | --- |
| 111L | 1 000-8 | 1 375-12UNF-2A | 1.300<br>1.290 | 1.015 | .100 | 1.290 | 1.225 | 1.500 | .103 | .070 | .250 | .550 |

(F) ENTIRE STANDARD REVISED

INCH-POUND

PREPARING ACTIVITY: ARMY-AR
CUSTODIANS: ARMY- AR NAVY- AS
AIR FORCE- 99 DLA-
REVIEW: AV, SH, 82, IS, NS
USER: AT, GL, MC
PROJECT NUMBER: 5340-2052

MILITARY SPECIFICATION SHEET

TITLE

INSERT, SCREW-THREAD, LOCKED IN, KEY-LOCKED, HEAVY DUTY

SPECIFICATION SHEET NUMBER
MS51831F 9 MAR 92

SUPERSEDING
MS51831E 6 SEPT 85

AMSC- N/A FSC- 5340

DISTRIBUTION STATEMENT

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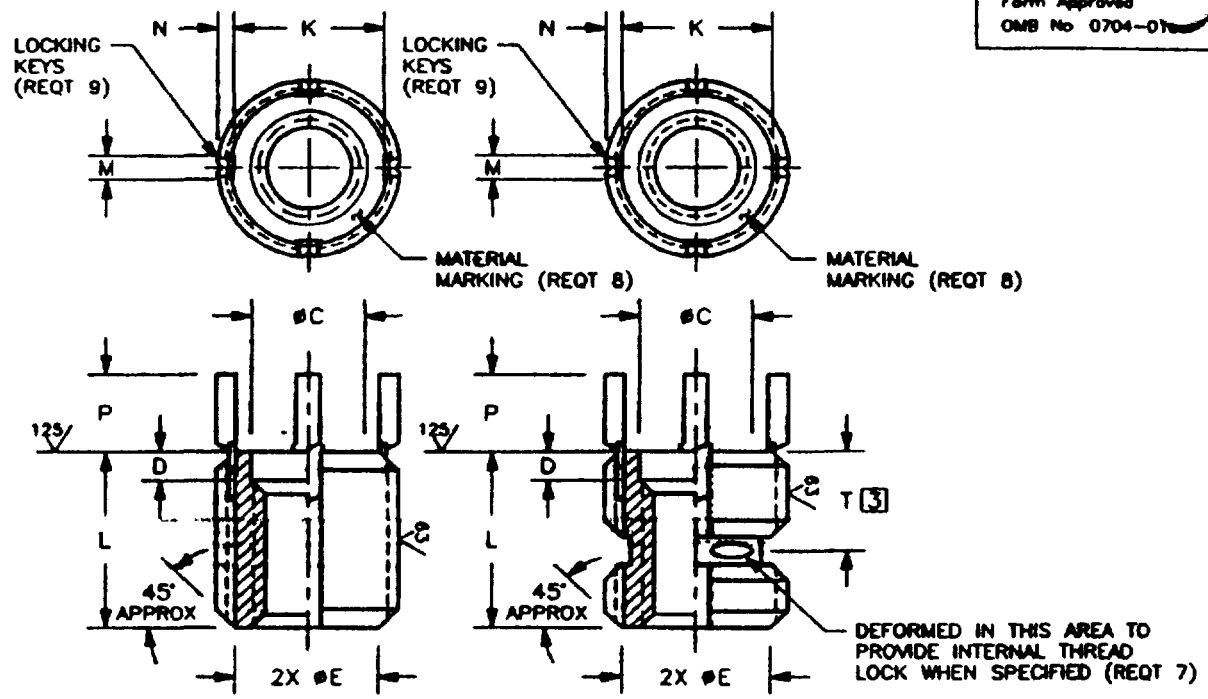



TABLE II FINE INTERNAL THREADS

| DASH NO. | INTERNAL THREAD MIL-S-8879 UNJF-3B | EXTERNAL THREAD FED-STD-H28/2 EXCEPT MODIFIED MINOR DIA. | | øC | D | øE | K | L | KEY DIMENSIONS | | | T |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SIZE | øMINOR | ±.005 | ±.010 | REF | REF | ±.015 | M REF | N REF | P REF | ±.030 |
| 201 | .190-32 | .375-16UNC-2A | .320<br>.312 | .196 | 070 | .310 | .284 | .312 | .068 | .040 | .130 | .220 |
| 202 | .250-28 | .4375-14UNC-2A | .383<br>.375 | .257 | 070 | .375 | .344 | .375 | .068 | .040 | .190 | .220 |
| 203 | 3125-24 | .500-13UNC-2A | .440<br>.432 | .316 | .070 | .430 | .407 | 437 | .068 | .040 | .190 | .220 |
| 204 | .375-24 | .5625-12UNC-2A | .503<br>.495 | .380 | .070 | .495 | 469 | .500 | .068 | .040 | .190 | .250 |
| 205 | 4375-20 | 625-11UNC-2A | .565<br>.551 | 445 | 100 | 550 | 532 | 625 | .068 | 040 | 190 | .280 |
| 206 | 500-20 | 6875-11NS-2A | .625<br>.615 | 507 | 100 | 615 | .595 | 688 | .068 | 040 | .190 | .280 |
| 207 | .5625-18 | .8125-16UN-2A | .752<br>.744 | .580 | 100 | 745 | 720 | 812 | .068 | 040 | .190 | .320 |
| 208 | .625-18 | .875-14UNF-2A | .815<br>.807 | .640 | .100 | .805 | .782 | .875 | .068 | .040 | .190 | .330 |
| 209 | .750-16 | 1.125-12UNF-2A | 1.050<br>1.040 | .763 | .100 | 1.040 | .975 | 1.125 | .103 | .070 | .250 | --- |
| 209L | .750-16 | 1.125-12UNF-2A | 1 050<br>1.040 | .763 | .100 | 1.040 | .975 | 1.250 | .103 | .070 | .250 | .490 |
| 210 | 875-14 | 1.250-12UNF-2A | 1 175<br>1 165 | .888 | .100 | 1.165 | 1.100 | 1.250 | .103 | .070 | .250 | --- |
| 210L | .875-14 | 1.250-12UNF-2A | 1.175<br>1 165 | .888 | 100 | 1.165 | 1 100 | 1.375 | 103 | .070 | 250 | .490 |
| 211 | 1.000-12 | 1.375-12UNF-2A | 1 300<br>1.290 | 1.015 | 100 | 1.290 | 1.225 | 1.375 | .103 | .070 | 250 | --- |
| 211L | 1.000-12 | 1.375-12UNF-2A | 1.300<br>1.290 | 1.015 | .100 | 1.290 | 1.225 | 1.500 | .103 | .070 | .250 | .550 |

PREPARING ACTIVITY: ARMY-AR

CUSTODIANS: ARMY- AR NAVY- AS
AIR FORCE- 99 DLA-

REVIEW: AV, SH, 82, IS, NS
USER: AT, GL, MC
PROJECT NUMBER: 5340-2052

MILITARY SPECIFICATION SHEET

TITLE

INSERT, SCREW-THREAD, LOCKED IN, KEY-LOCKED, HEAVY DUTY

SPECIFICATION SHEET NUMBER
MS51831F 9 MAR 92

SUPERSEDING
MS51831E 6 SEPT 85

AMSC- N/A FSC- 5340

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THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

Form Approved
OMB No 0704-0188

REQUIREMENTS

1 MATERIAL. STEEL, CORROSION-RESISTANT, TYPE 303 (UNS S30300) IN ACCORDANCE WITH AMS 5640 (TYPE 1) OR ASTM A 582, OR 303SE (UNS S30323) IN ACCORDANCE WITH AMS 5640 (TYPE 2), AMS 5738 OR ASTM A 582
STEEL, CORROSION-RESISTANT, TYPE A286 (UNS S66286) IN ACCORDANCE WITH AMS 5734 OR AMS 5737
STEEL, ALLOY, GRADE 4140 (UNS G41400) IN ACCORDANCE WITH MIL-S-5626 OR GRADE 8740 (UNS G87400) IN ACCORDANCE WITH AMS 6322
LOCKING KEYS STEEL, CORROSION-RESISTANT, TYPE 302 CHEMICAL COMPOSITION OF ASTM A 580 ONLY

2 CADMIUM PLATING AND SURFACE TREATMENT. CORROSION-RESISTANT STEEL SHALL BE PASSIVATED IN ACCORDANCE WITH QQ-P-35
ALLOY STEEL SHALL BE CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3 THE LOCKING KEYS MAY OR MAY NOT BE CADMIUM PLATED

3 LUBRICATION: INSERTS WITH SELF-LOCKING INTERNAL THREADS SHALL BE DRY FILM LUBRICATED IN ACCORDANCE WITH MIL-L-46010, TYPE I. THE LOCKING KEYS MAY OR MAY NOT BE LUBRICATED

4 SURFACE TEXTURE. MACHINED SURFACES SHALL BE IN ACCORDANCE WITH ANSI/ASME B46 1

5 HEAT TREATMENT. ALLOY STEEL INSERTS SHALL BE HEAT TREATED TO 160,000 psi Ftu MINIMUM IN ACCORDANCE WITH MIL-H-6875
CORROSION-RESISTANT STEEL INSERTS, TYPE A286 (AMS 5734), SHALL BE HEAT TREATED TO 140,000 psi Ftu MINIMUM

6. HARDNESS. ALLOY STEEL INSERTS SHALL HAVE A HARDNESS RANGE OF 36-40 HRC
CORROSION-RESISTANT STEEL INSERTS, TYPE A286, SHALL HAVE A HARDNESS RANGE OF 29-38 HRC.

7. PART NUMBER. THE PART NUMBER SHALL CONSIST OF THE BASIC MS SHEET NUMBER PLUS THE DASH NUMBER TAKEN FROM TABLE I OR TABLE II, AS APPLICABLE
EXAMPLE. MS51831 - 102 L
- ADD "L" AS SUFFIX TO DASH NUMBER FOR INTERNAL THREAD LOCK LEAVE BLANK IF INTERNAL THREAD LOCK IS NOT REQUIRED
- DASH NUMBER FROM TABLE I OR TABLE II
- MATERIAL. DASH INDICATES CRES, TYPE 303 OR 303SE; ADD "CA" IN LIEU OF DASH FOR CRES, TYPE A286; ADD "A" IN LIEU OF DASH FOR ALLOY STEEL, GRADE 4140 OR 8740
- BASIC MS SHEET NUMBER

EXAMPLE. MS51831CA102L INDICATES - INSERT, SCREW-THREAD, LOCKED IN, KEY-LOCKED, HEAVY DUTY, CRES A286, .250-20UNJC-3B INTERNAL THREAD WITH INTERNAL THREAD LOCK FEATURE

8. MATERIAL MARKING. CRES 303 OR 303SE HAS NO IDENTIFYING MARK.
CRES A286 SHALL BE IDENTIFIED ON TOP OF INSERT BY ONE (1) LINE OR DASH MARK.
ALLOY STEEL 4140 OR 8740 SHALL BE IDENTIFIED ON TOP OF INSERT BY TWO (2) PARALLEL LINES OR DASH MARKS

9 INSERTS WITH INTERNAL THREAD SIZE 250 AND SMALLER SHALL BE SUPPLIED WITH TWO (2) LOCKING KEYS SPACED 180° APART
INSERTS WITH INTERNAL THREAD SIZE 3125 AND GREATER SHALL BE SUPPLIED WITH FOUR (4) LOCKING KEYS SPACED 90° APART

10 INSERTS SHALL BE FREE OF ALL HANGING BURRS AND SLIVERS WHICH MIGHT BECOME DISLODGED UNDER USAGE

11 SOURCE IDENTIFICATION MARK. SOURCE IDENTIFICATION MARK SHALL BE IN ACCORDANCE WITH MIL-I-45914

12. ALL DIMENSIONS ARE AFTER CADMIUM PLATING OR SURFACE TREATMENT AND PRIOR TO THE ADDITION OF THE LUBRICATION

13. FILLETS ARE R.015 MAXIMUM.

NOTES

1. ALL DIMENSIONS ARE IN INCHES

2. INSTALLATION OF INSERTS SHALL BE IN ACCORDANCE WITH MS51835

[3] DISTANCE TO CENTER OF INTERNAL THREAD LOCK.

4 IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

5 REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

| | | |
|---|---|---|
| PREPARING ACTIVITY: ARMY-AR<br>CUSTODIANS: ARMY- AR NAVY- AS<br>AIR FORCE- 99 DLA-<br>REVIEW: AV, SH, 82, IS, NS<br>USER: AT, GL, MC<br>PROJECT NUMBER 5340-2052 | MILITARY SPECIFICATION SHEET<br>TITLE<br>INSERT, SCREW-THREAD, LOCKED IN, KEY-LOCKED, HEAVY DUTY | SPECIFICATION SHEET NUMBER<br>MS51831F 9 MAR 92<br>SUPERSEDING<br>MS51831E 6 SEPT 85<br>AMSC- N/A FSC- 5340 |
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